1. Use 8086 Interfacing Trainer in Kit mode to make a traffic controlling system with three LEDs as the represententing light of Red, Green, Yellow color light. Here the lights will be displayed in the following sequence and time period:

1. Red light will be on for 10 seconds
2. Yellow light will be on for 5 seconds
3. Green light will be on for 15 seconds
4. Red light will be on for 10 seconds
5. Repeat the sequence

Code:

BO, 80	FE, CA	FE, CA
E6, 1F	75, F7	75, F7
BO, FF	B0, F4	B0, F4
E6, 1B	E6, 1B	E6, 1B
E6, 19	B2, 05	B2, 05
B0, 00	B9, FF, FF	B9, FF, FF
E6, 1D	E2, FE	E2, FE
B0, F1	FE, CA	75, F7
E6, 1B	75, F7	EB, C2
B2, 0A	B0, F2	
B9, FF, FF	E6, 1B	
E2, FE	B2, 0F	
	B9, FF, FF	
	E2, FE	

2. Use 8086 Interfacing Trainer in Kit mode to display letters E, F and H on a 7-Segment LED with a delay 5 seconds between each display. The output will be continued until the system is reset.

CODE:

B0, 80	B2, FF	B2, FF
E6, 1E	B0, 00	B0, 00
B0, FF	E6, 1A	E6, 1A
E6, 18	B0, 03	B0, 03
B2, FF	E6, 1C	E6, 1C
B0, 00	B9, FF, 01	B9, FF, 01
E6, 1A	E2, FE	E2, FE
B0, C3	B0, 27	B0, 24
E6, 1C	E6, 1A	E6, 1A
B9, FF, 01	BO, FF	BO, FF
E2, FE	E6, 1C	E6, 1C
B0, E7	B9, FF, 01	B9, FF, 01
E6, 1A	FE, CA	FE, CA
BO, FF	75, E2	75, E2
E6, 1C		EB, 9E
B9, FF, 01		
FE, CA		
75, E2		

3. Use 8086 Interfacing Trainer in Kit mode to display digits 1, 2 and 3 on a 7-Segment LED with a delay 5 seconds between each display. The output will be continued until the system is reset.

CODE:

B0, 80	FE, CA
E6, 1F	75, F7
B0, F0	B0, B0
E6, 1B	E6, 19
B0, F9	B2, 05
E6, 19	B9, FF, FF
B2, 05	E2, FE
B9, FF, FF	FE, CA
E2, FE	75, F7
FE, CA	EB, D1
75, F7	
B0, A4	
E6, 19	
B2, 05	
B9, FF, FF	
E2, FE	

4. Use 8086 Interfacing Trainer in Kit mode to display a vertical line of LEDs in an 8*8 dot matrix LEDs and then shift the display from left to right with a delay 5 seconds between each display. The output will be continued until the system is reset.

CODE:

```
BO, 80
E6, 1E
BO, FF
E6, 18
B2, 01
BO, 00
E6, 1A
8A, C2
02, D2
E6, 1C
B9, FF, FF
E2, FE
72, ED
EB, ED
```

5. Use 8086 Interfacing Trainer in Kit mode to display a horizontal line of LEDs in an 8*8 dot matrix LEDs and then shift the display from top to Bottom with a delay 5 seconds between each display. The output will be continued until the system is reset.

CODE:

```
BO, 80
E6, 1E
BO, FF
E6, 18
B2, 80
BO, FF
E6, 1C
32, C2
D0, EA
E6, 1A
B9, FF, FF
E2, FE
72, ED
EB, ED
```